

ENEL301 – Fundamentals of Engineering Economics and Management

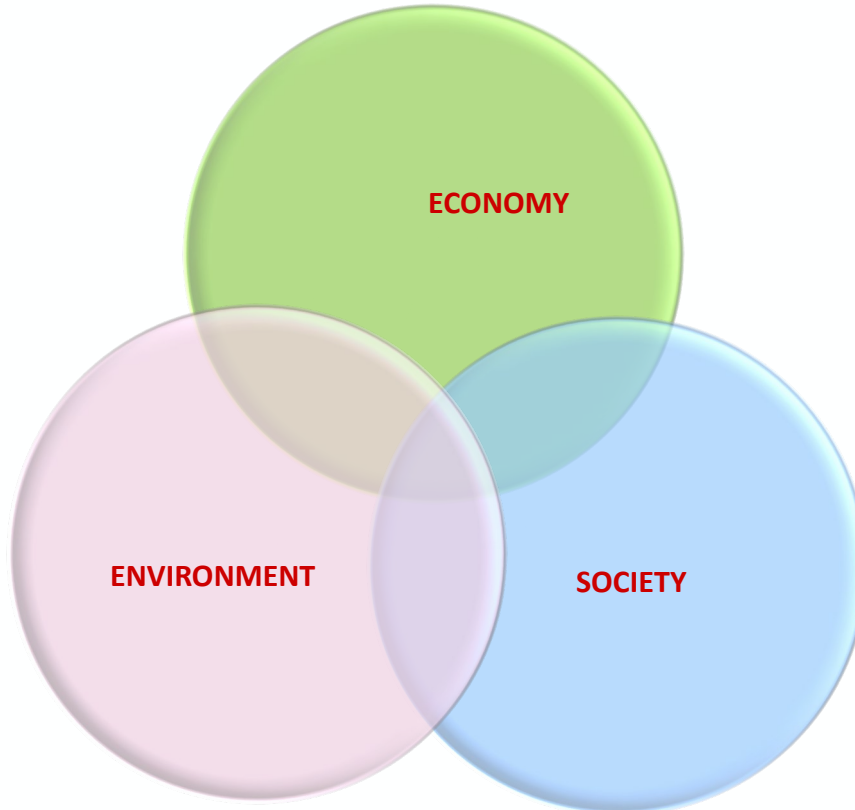
Sustainability

Life Cycle Assessment (1 of 2)

What we are going to cover and when...

Week	Lectures Tuesday 10-11 & Thursday 9-10	Tutorial / Workshop Weeks 1, 4-6 Wednesday 3-4 or 4-5. Weeks 2&3 Wednesday 3-5
1	Introduction to Engineering Practice & Ethics	Ethics
	Collaboration & Teamwork	
2	Māori Perspectives in Engineering	Bicultural Competence and Confidence (timetable and attend one of B01-07 OR B08-14)
	Bias, Diversity & Ethics	
3	Intellectual Property (1/2)	
	Intellectual Property (2/2)	
4	Law in New Zealand (1/2)	IP Protection
	Law in New Zealand (2/2)	
5	Data Privacy and Data Ethics	Law, Data Privacy, Data Ethics
	Business Structures, Strategy, Decisions & Risk	
6	Sustainable Development & Systems Thinking	Sustainable Development & Systems Thinking
	Life Cycle Assessment (1/2)	

What is sustainability?



“..meeting the needs of the present without compromising the ability of future generations to meet their own needs.”

Brundtland Report – Our Common Future



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Some more about my background

Companies I've completed LCAs for...



Life Cycle Assessment

- A tool to evaluate the environmental performance of a product or service
 - Society, **Environment**, Economy
- Framework: ISO 14040:2006 and ISO 14044:2006
- Similar approach can be used to assess social impacts (Social LCA)
- NOT life cycle costing (although LCC can be done in parallel with LCA)
- “Classic” LCA questions
 - Drinking containers: Glass vs. PET vs. Aluminium
 - Disposable vs cotton nappies
 - Electric car vs petrol car
 - Does the embodied energy in a solar panel ever pay for itself?

Why is this important?

Your value creation

Design

1. Innovation, research and development
2. Product differentiation

Engineering Analysis:

3. Efficiency improvement
4. Uncertainty reduction
5. Performance forecasts

Due diligence:

6. Inspection, Checking, Testing
7. Project and Design Review
8. Compliance with standards

Collaboration

9. Reliable coordination of other people (who then create value)
10. Teaching important skills & knowledge
11. Co-creating value with communities

Sustainment

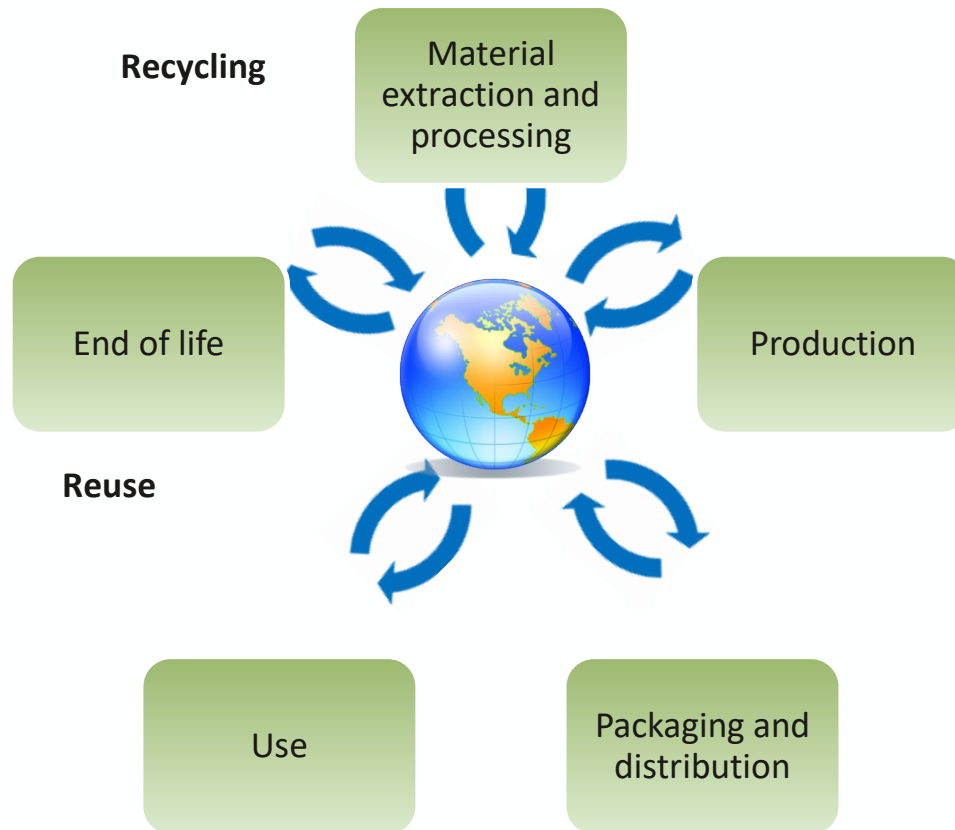
12. Operations, asset management and maintenance

Protection

13. Environmental Protection
14. Security & Defence

Life Cycle Assessment

The basics



Quantify the environmental impacts over the full life cycle.... Product or service

Impact category

Global warming (Carbon)

Embodied energy

Acidification

Eutrophication

Ecological footprint

Water use

Carcinogens

Typical units

kg CO₂-eq

MJ

kg SO₂-eq

kg PO₄⁻³-eq

Ha.a

kL

DALY

Resources



Emissions

Life Cycle Assessment

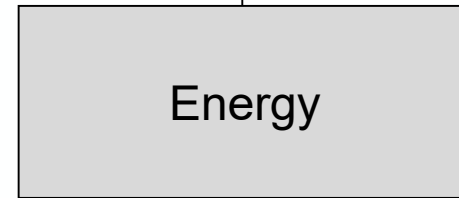
The basics

Emissions to air,
water and ground (e.g. NO_x)



Raw materials and resources
(e.g. iron ore, land, water)

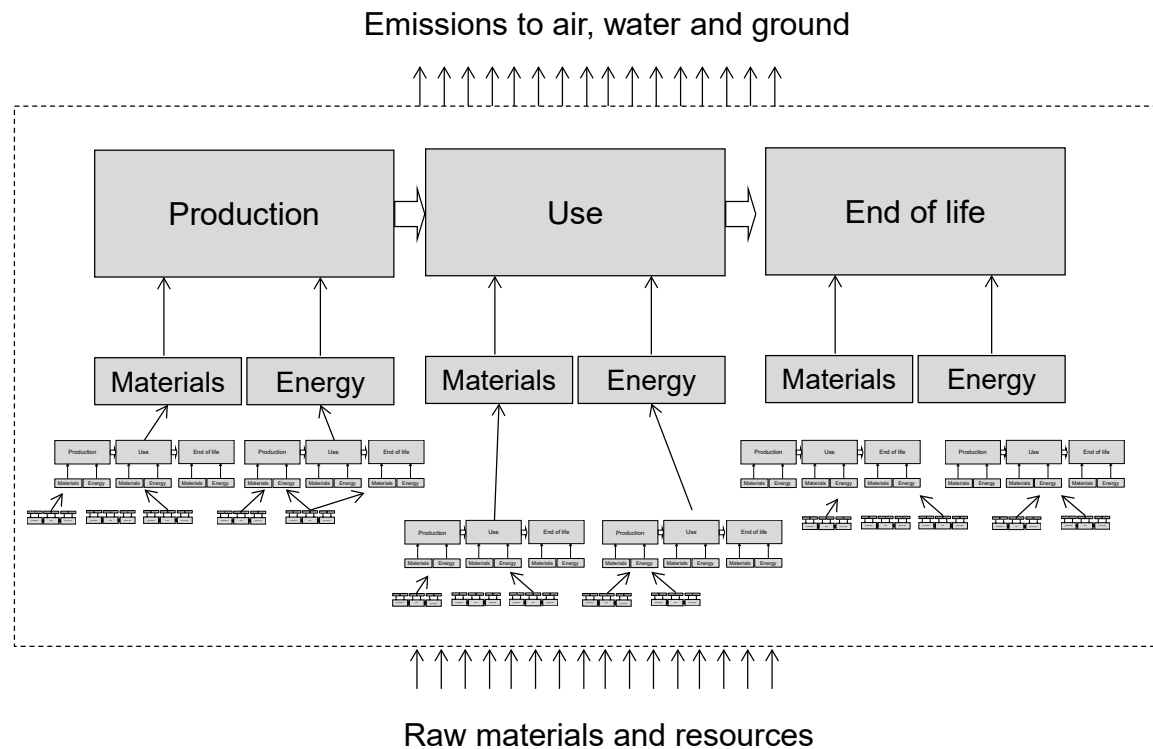
Emissions to air,
water and ground (e.g. CO₂)



Raw materials and resources
(e.g. natural gas)

Life Cycle Assessment

The basics

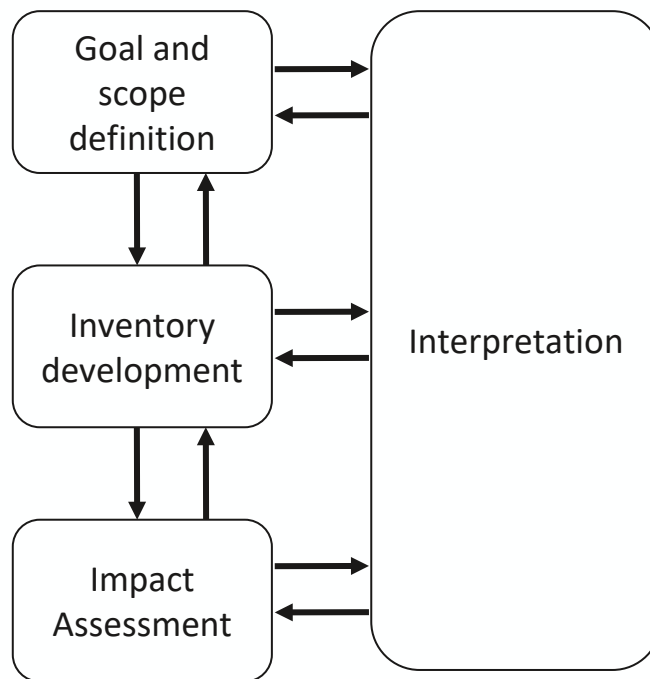


Life Cycle Assessment

Reasons why

- Product development and improvement
- Marketing
- Design tool
- Public policy development
- Strategic planning
- Entry to market (e.g. for building products)

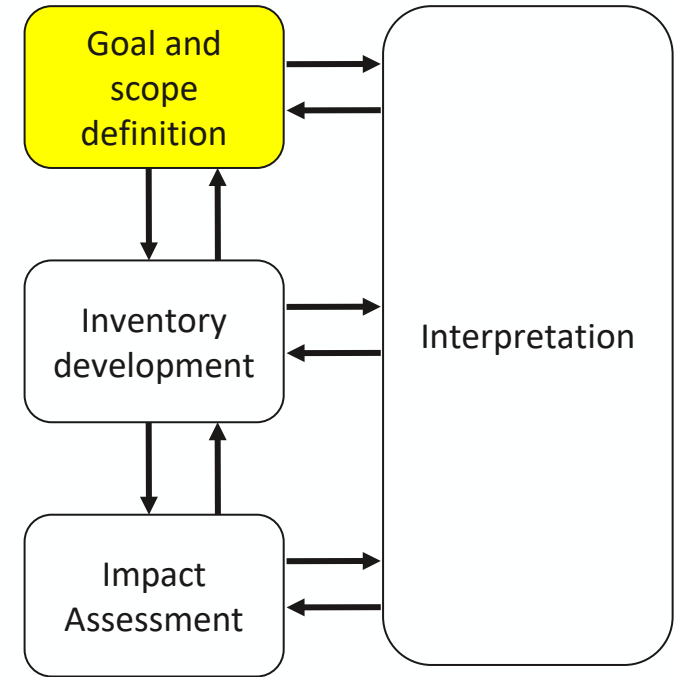
How to do an LCA



Source: ISO 14040:2006

Goal and scope definition

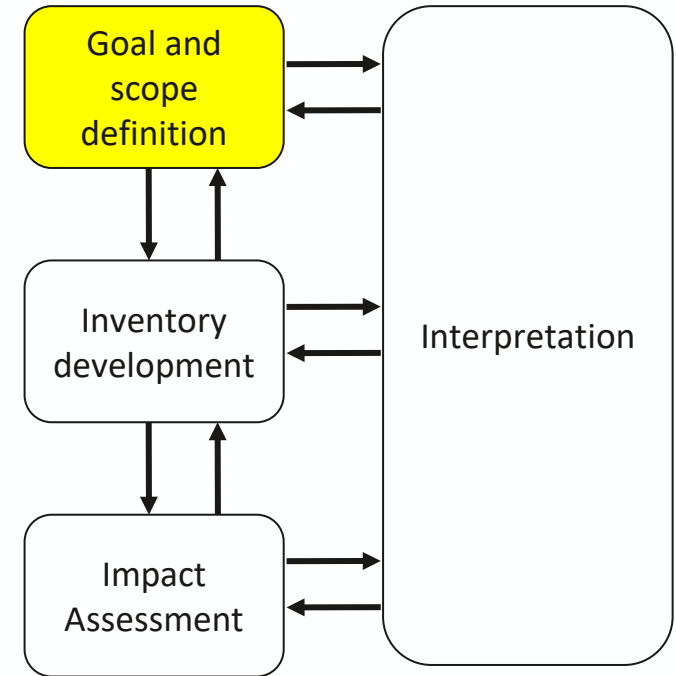
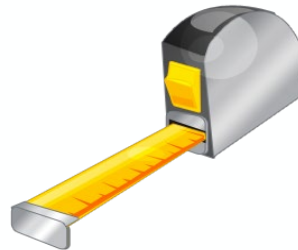
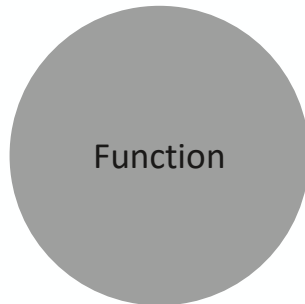
- Frame the study
 - Reasons for study & objectives
 - Audience, e.g. internal, external, marketing purpose
- Key methodology, including:
 - Functional unit
 - System boundary
 - What is included/excluded
 - Identify environmental indicators
 - Identify what data is needed



Source: ISO 14040:2006

Functional unit

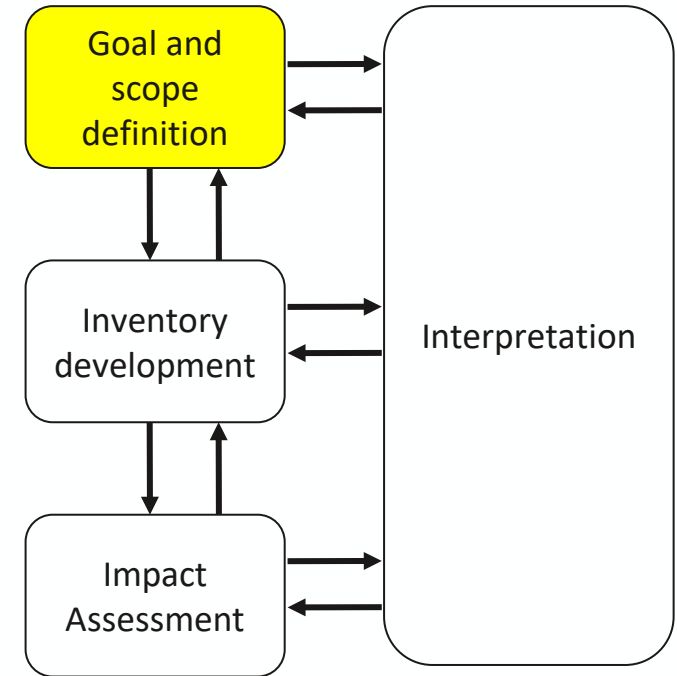
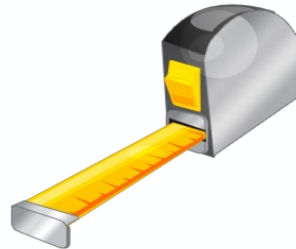
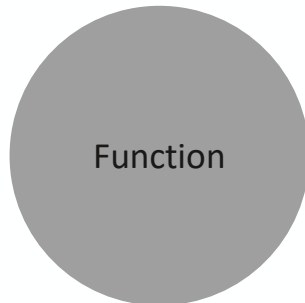
- Identify the primary function
- Quantify that function
- Allows for fairer comparison between systems



Source: ISO 14040:2006

Functional unit

- Soft-drink packaging: Glass vs. PET vs. Al
 - Delivery of 1 L of soft drink to consumer
- Electric car vs petrol car
 - Move 1 person one kilometre (person.km)
- Does the embodied energy in a solar panel ever pay for itself?
- 1 kWh of electricity delivered to a household



Source: ISO 14040:2006

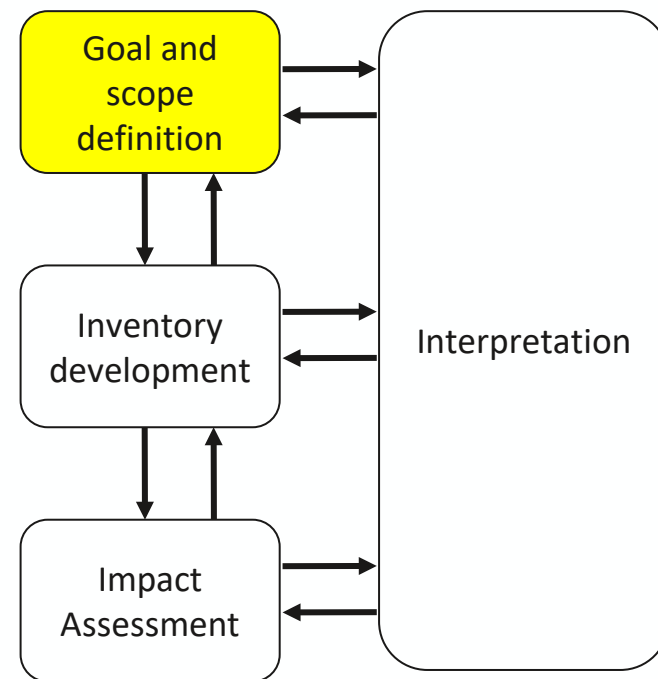
System boundary

“...which ... processes are part of a product system” (ISO 14040:2006)

- Attempt to cover 95% + of environmental impacts associated with the product life cycle
- Key challenges is to know what is missing
 - Reviewing guidelines & previous studies helps

Typical exclusions:

- Capital equipment and maintenance
- Human labour, except for labour-intensive
- Accidents



Source: ISO 14040:2006

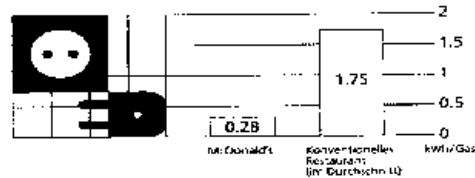
System Boundary

The McDonald's example

McDonald's ist umweltbewusst!

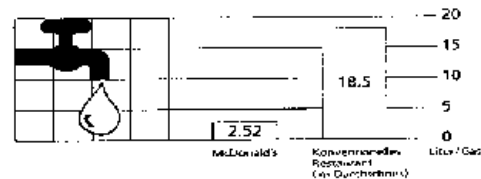
Der Beweis:

6 × weniger Energie!



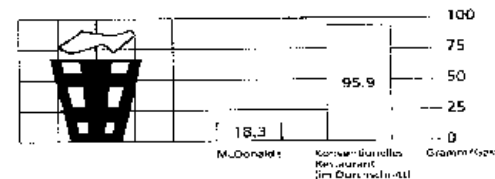
6 x less energy

7 × weniger Wasser!



7 x less water

5 × weniger Abfall!

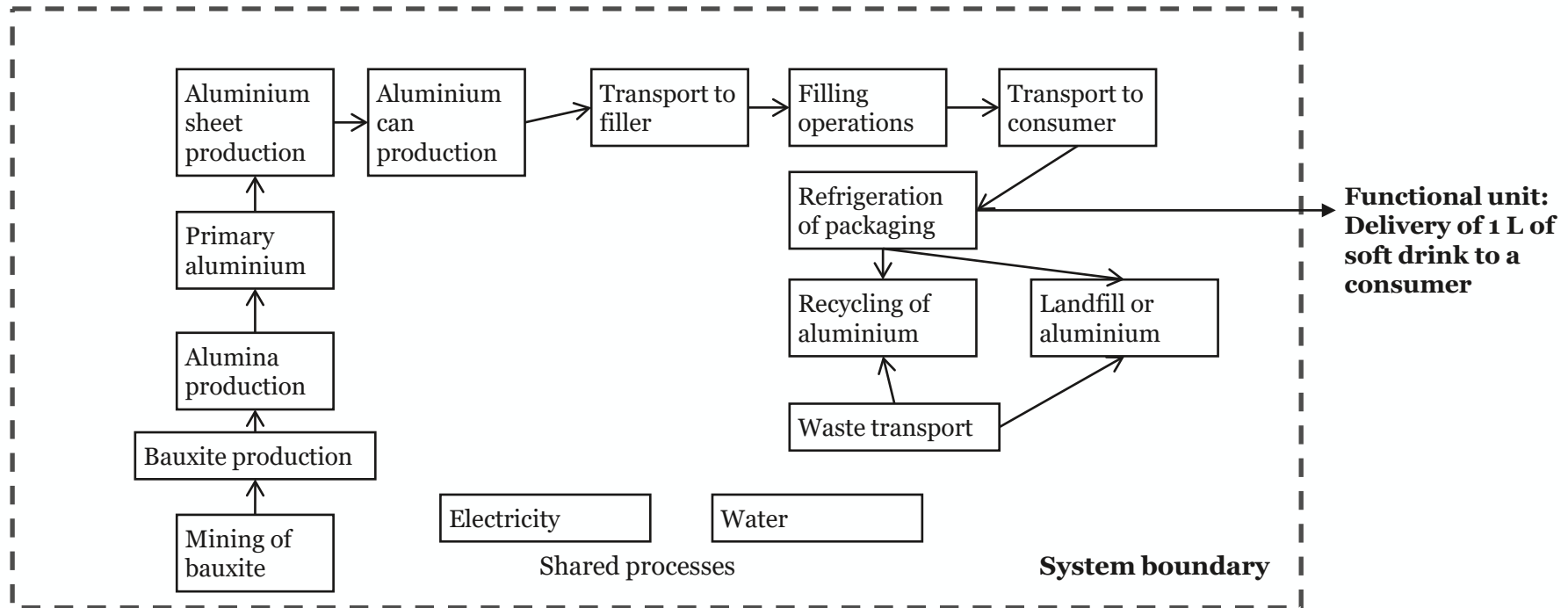


5 x less waste

compared to a
conventional restaurant

System Boundary – Classic LCA questions

Soft-drink packaging: Glass vs. PET vs. Al



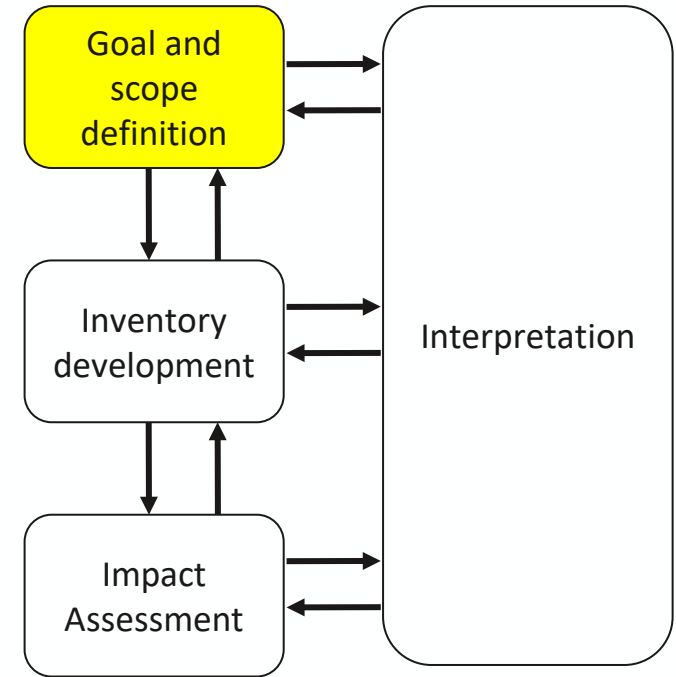
**Outside of
boundary
(exclusions)**

Identify environmental indicators

- What environmental indicators are you interested in studying?
- Need to be relevant
- Do not ignore known issues
- 30+ indicators you could consider

Where to start:

- What do stakeholders think are important?
- What indicators have been used elsewhere?
 - Search for similar studies → Google or Google Scholar (ChatGPT will likely hallucinate, or convince you that you are right)
- Guidelines / related standards

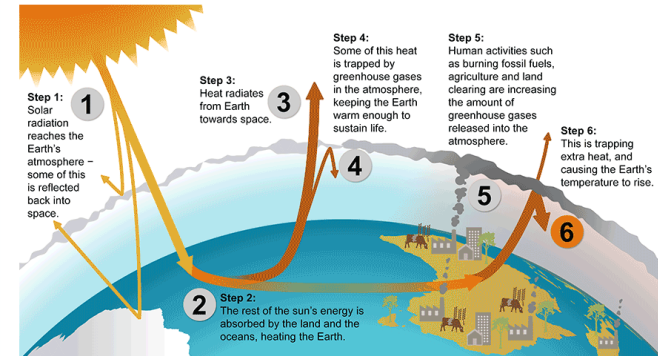


Source: ISO 14040:2006

Identify environmental indicators

Climate change potential

- Air emissions of CO₂, CH₄, N₂O, SF₆, CFCs etc
- Often tracks energy from fossil fuels, so can be a good proxy for many other fossil fuel related impacts
- *Climate change potential, kg CO₂-eq*
- Carbon accounting methods, inc. time horizons can affect result



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Identify environmental indicators

Smog

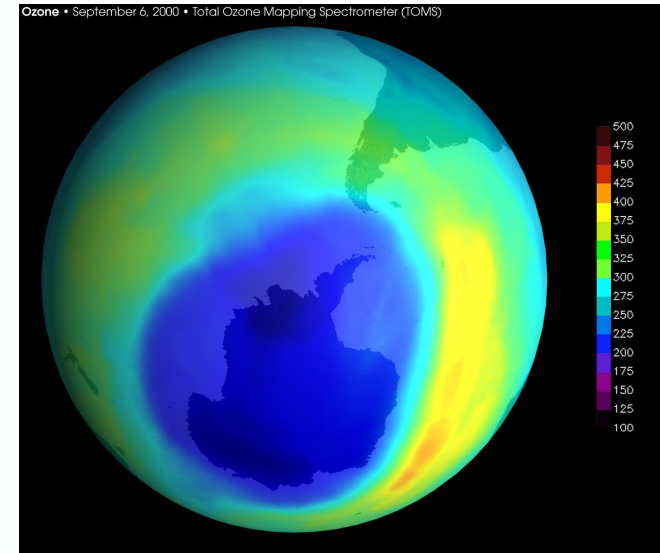
- Air emissions of substances which contribute to smog formation, e.g. Nox, VOCs etc
- *Smog potential, kg O_3 - eq*



Identify environmental indicators

Ozone depletion

- Air emissions of ozone depleting substances
- *Ozone depletion potential, kg CFC-11 eq*



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Identify environmental indicators

Acidification

- Air, soil and water emissions of acidifying substances, e.g. SO_x, NO_x, HF, etc. etc.
- Acid rain is a problem mainly in Scandinavia, eastern Europe, east United States, SE Canada, Taiwan, SE coast of China
- *Acidification potential*, kg SO₂-eq



Identify environmental indicators

Eutrophication/Nutrification

- Air, soil and water emissions of NO_x, PO_x etc. etc.
- Eutrophication of aquatic and nutrification of terrestrial ecosystems is caused by surplus nitrogen, phosphorus, and degradable organic substances. Manifests in algal blooms (in freshwater)
- *Eutrophication Potential*, kg PO₄³⁻-eq (phosphate equivalents) OR kg N-eq (nitrogen equivalents)

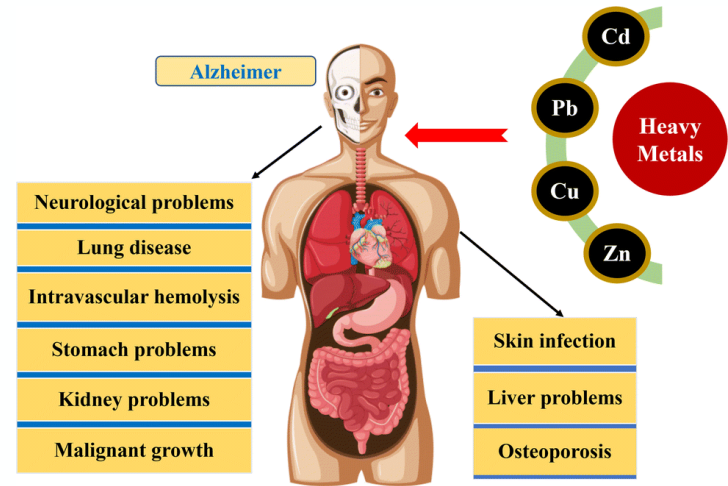


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Identify environmental indicators

Human toxicity

- Air, soil and water emissions of dioxins, cadmium, arsenic etc. etc.
- Can be sub-classified into carcinogens and non-carcinogens
- Comparative Toxic Unit for human (CTUh)
 - estimated increase in morbidity in the total human population per unit mass of a chemical emitted



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Identify environmental indicators

Eco-toxicity

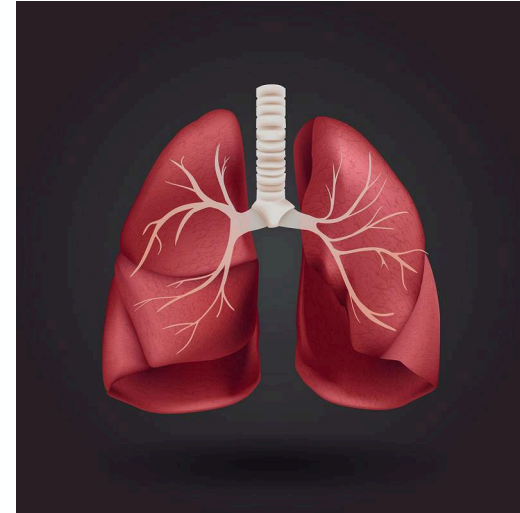
- Air, soil and water emissions of dioxins, cadmium, arsenic etc. etc.
- Comparative Toxic Unit for ecosystems (CTUh)
 - Like CTUh, but for everything else



Identify environmental indicators

Respiratory effects

- Air emissions of small particulate matter
- Some overlap with human health indicators
- *Respiratory effects kg PM_{2.5}-eq*



Identify environmental indicators

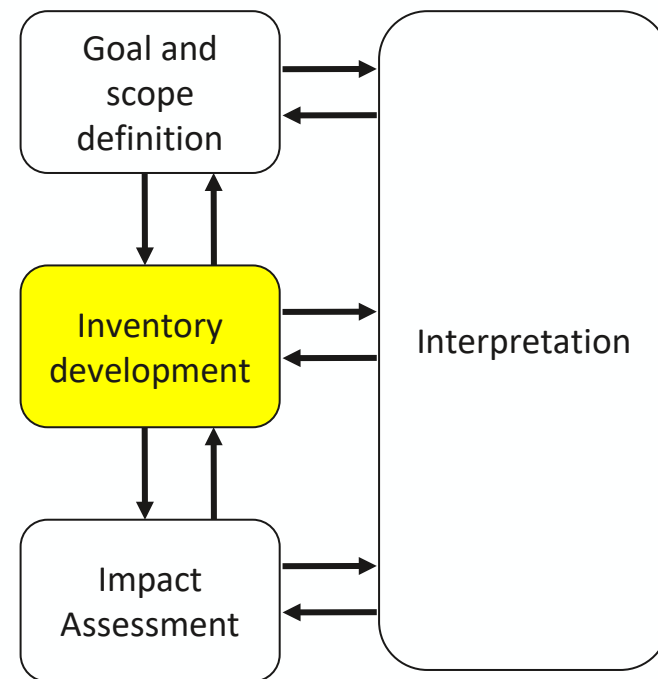
Minerals and fossil fuel depletion

- Consumption of minerals and fossil fuels
- Measure of the pressure / strain on resource use
- *Fossil fuel or Minerals depletion*
 - Additional energy (kg MJ surplus)
 - Additional investment (\$)



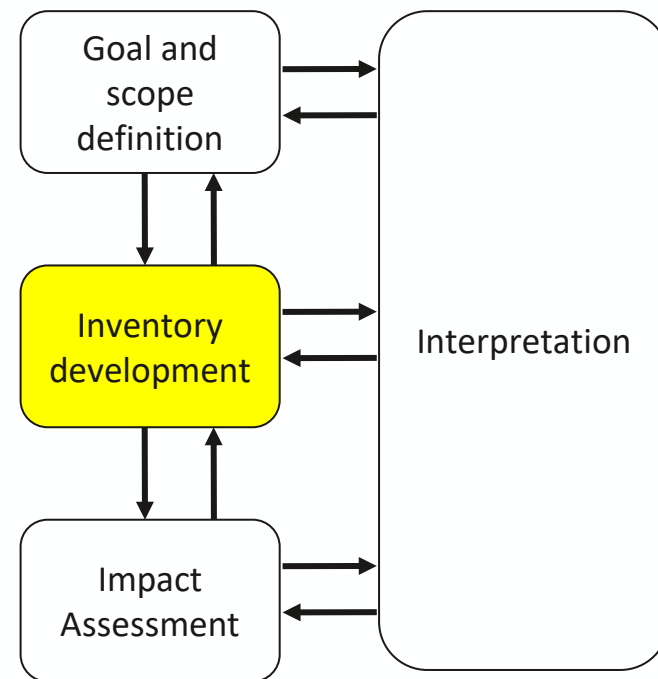
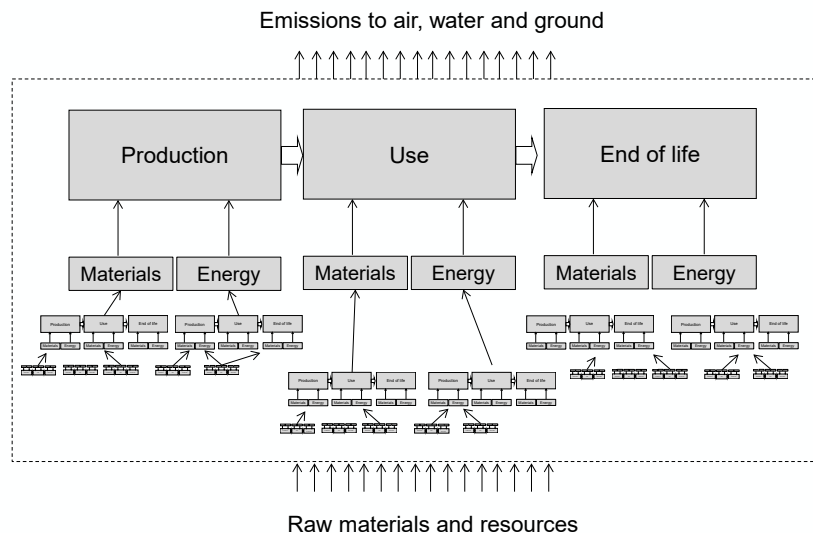
Inventory development

- LCA projects rely on excellent data
 - But there's no such thing as 'perfect'
- Foreground data
 - Data that the original manufacturer supplies, e.g. mass of packaging, energy inputs for a process, emissions
- Background data
 - Data that the LCA practitioner supplies, e.g. electricity generation, transport, production of common materials
 - Typically from LCA databases
- Data cut-off at 5% (e.g. by mass, impact)



Source: ISO 14040:2006

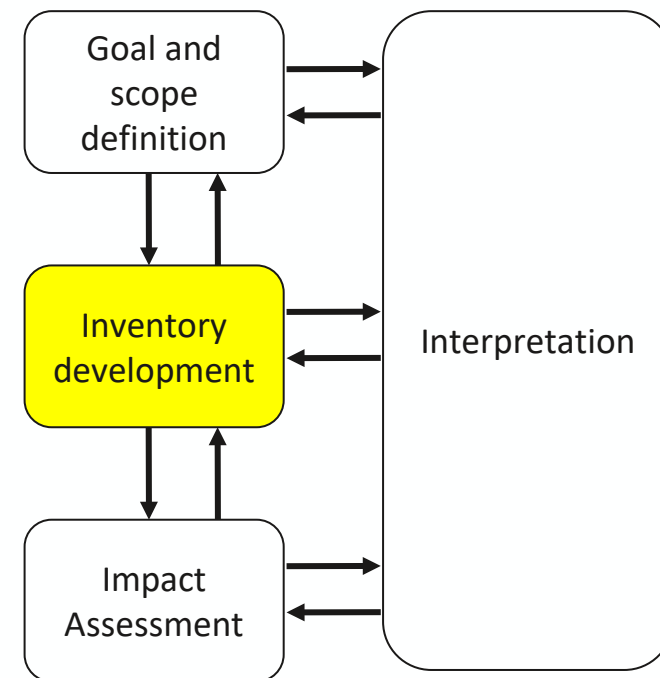
Inventory development



Source: ISO 14040:2006

Inventory development

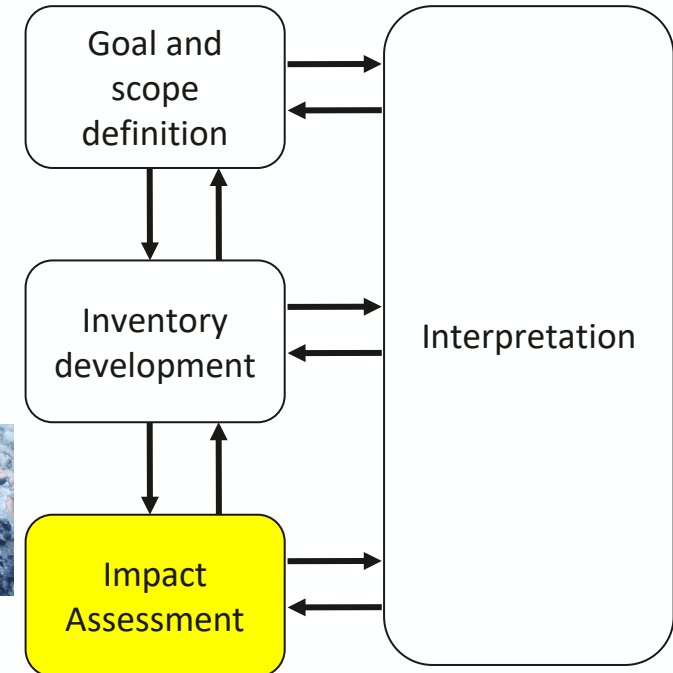
Substance	Compartment	Per functional unit		
		Product A	Product B	Unit
Inflows				
Additives	Raw	2.94	0.43	µg
Air	Raw	406.22	180.04	ng
Aluminum hydroxide	Raw	168.16	180.16	mg
Carbon	Raw	1.28	0.57	mg
Coal washery waste	Raw	0.00	56.95	g
Energy, from solar	Raw	5.76	2.55	kJ
Energy, kinetic, flow, in wind	Raw	0.00	193.63	J
Energy, recovered	Raw	0.00	0.00	J
Energy, unspecified	Raw	369.93	106.59	MJ
Ferromanganese	Raw	505.11	73.32	µg
Granite, in ground	Raw	1.14	0.00	tonne
Graphite, from technosphere, 50MJ/kg	Raw	221.81	32.20	ng
Occupation, industrial area	Raw	0.61	0.29	m2s
Phosphorus pentoxide	Raw	77.64	82.99	mg
Potassium chloride	Raw	12.23	5.42	ng
Secondary glass	Raw	115.43	19.29	µg
Sulfur dioxide, secondary	Raw	321.99	0.25	g
Waste, concrete	Raw	-1.44	-1.66	tonne
Waste, fly ash	Raw	75.00	170.00	kg
bagasse	Raw	29.39	13.57	g
bagasse, 8.7MJ/kg	Raw	572.88	83.16	mg
Energy, from biomass	Raw	4.53	2.01	MJ
etc. etc.				
Outflows				
Carbon dioxide	Air	269.79	230.95	kg
Carbon dioxide, biogenic	Air	256.60	114.33	g
Carbon dioxide, fossil	Air	50.43	22.45	kg
Carbon disulfide	Air	0.80	3.52	mg
Carbon monoxide	Air	379.78	451.32	g
Carbon monoxide, biogenic	Air	0.00	100.41	mg
Carbon monoxide, fossil	Air	0.00	1.42	g
Ethane, 1,1,1-trichloro-, HCFC-140	Air	0.00	0.01	pg
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	Air	21.37	45.17	mg
Ethane, 1,2-dibromo-	Air	28.27	33.20	µg
Ethane, 1,2-dichloro-	Air	640.86	284.03	pg
1-Methylfluorene	Water	6.71	7.87	µg
2-Hexanone	Water	384.10	451.09	µg
2-Methylnaphthalene	Water	0.93	1.09	mg



Source: ISO 14040:2006

Impact assessment

Substance	Compartment	Per functional unit		
		Product A	Product B	Unit
Inflows				
Additives	Raw	2.94		0.43 µg
Air	Raw	406.22		180.04 ng
Aluminum hydroxide	Raw	168.16		180.16 mg
Carbon	Raw	1.28		0.57 mg
Coal washery waste	Raw	0.00		56.95 g
Energy, from solar	Raw	5.76		2.55 kJ
Energy, kinetic, flow, in wind	Raw	0.00		193.63 J
Energy, recovered	Raw	0.00		0.00 J
Energy, unspecified	Raw	369.93		106.59 MJ
Ferromanganese	Raw	505.11		73.32 µg
Granite, in ground	Raw	1.14		0.00 tonne
Graphite, from technosphere, 50MJ/kg	Raw	221.81		32.20 ng
Occupation, industrial area	Raw	0.61		0.29 m2s
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etc. etc.				
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Carbon dioxide, biogenic	Air	256.60		114.33 g
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Carbon disulfide	Air	0.80		3.52 mg
Carbon monoxide	Air	379.78		451.32 g
Carbon monoxide, biogenic	Air	0.00		100.41 mg
Carbon monoxide, fossil	Air	0.00		1.42 g
Ethane, 1,1,1-trichloro-, HCFC-140	Air	0.00		0.01 pg
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	Air	21.37		45.17 mg
Ethane, 1,2-dibromo-	Air	28.27		33.20 µg
Ethane, 1,2-dichloro-	Air	640.86		284.03 pg
1-Methylfluorene	Water	6.71		7.87 µg
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Source: ISO 14040:2006

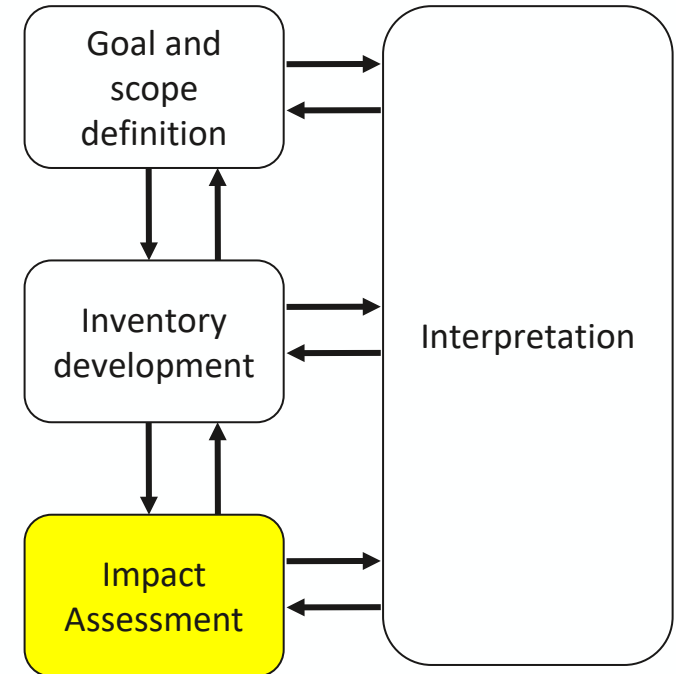
Impact assessment

Must:

- Characterise: calculate impact assessment
- Typically 'mid-point' in NZ, but 'end-point' more common in Europe

Optional:

- Normalise: contribution relative to a known baseline (e.g. % of national emissions)
- Weighting of indicators to fewer scores
- Subjective → need to calculate for different weightings



Source: ISO 14040:2006

Impact assessment - characterisation

Air emission	Flow kg emitted / functional unit	Characterisation factor kg CO ₂ -eq/ kg emitted	Characterised result kg CO ₂ -eq / functional unit
CO ₂	0.1	1	0.1 x 1
CH ₄	0.09	21	0.09 x 21
N ₂ O	0.00012	310	0.00012 x 310
Sum			2.0272

Impact assessment - characterisation

Inventory

Compartment	Elementary flow	Amount (kg/ functional unit)
Raw	Air	0.01563
Raw	Aluminium, 24% in bauxite, 11% in crude ore, in ground	2.11E-02
Raw	Aluminum hydroxide	2.11E-02
Air	Cyclohexane	3.91E-08
Air	Diethylamine	1.47E-20
Air	Dimethyl malonate	7.86E-22
Air	Dinitrogen monoxide	2.35E-06
etc.		

X

Indicator 1

Compartment	Elementary flow	Characterisation factor (\$/kg)
Raw	Coal, 13.3 MJ per kg, in ground	0.01563
Raw	Coal, 18 MJ per kg, in ground	2.11E-02
Raw	Coal, 18.0 MJ per kg, in ground	2.11E-02
etc.		

Indicator 2

Compartment	Elementary flow	Characterisation factor (kg NMVOC/kg)
Air	1-Butanol	1.05
Air	1-Butene	1.82
Air	1-Butene, 2-methyl-	1.3
etc.		

X

Interpretation

Iterate back to other steps to:

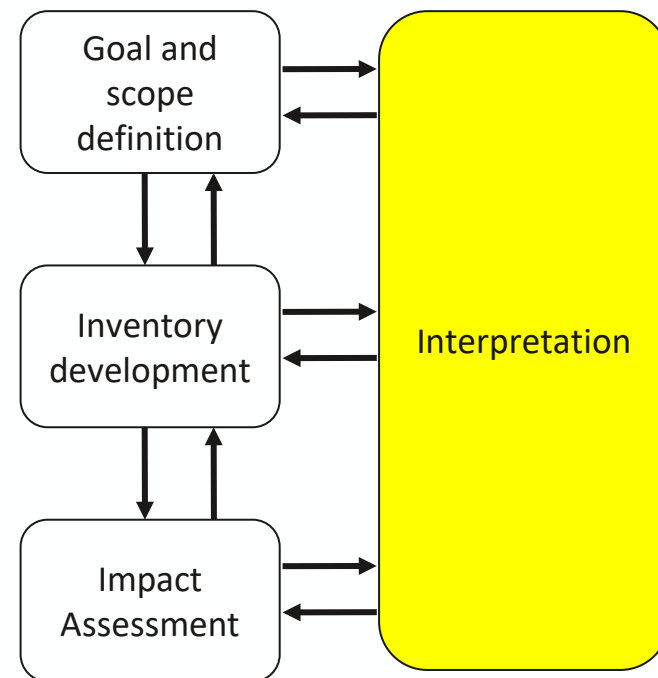
- Identify the reasons for your results → answer “Why?” 5 times
- Check methods inc. assumptions, data quality

Sensitivity studies

- The effect of key assumptions on your outcomes

Uncertainty studies

- The effect of data uncertainty on outcomes → confidence in your claims
- Monte Carlo simulations



Source: ISO 14040:2006

Practicalities of LCA

Factor	Streamlined LCA Basic	Streamlined LCA with tool-support	Full LCA
Typical use	Get a feel for the issues	Quick decisions, single product or comparisons	Detailed decisions or comparative claims
Standard compliance	None	Typically aligned to ISO 14040/14044	ISO 14040 / 14044 (or other international standard)
Data	Generic, publicly available	Generic, limited dataset	Supply-chain specific
Software	Spreadsheets	Online or integrated tools	OpenLCA, Sphera (GaBi), SimaPro
Resolution	Low	Medium	High to very high
Uncertainty	Very high	Medium to High	Low
Expertise level needed	Low	Medium to high	High to very high
Time taken	Hours	Days to weeks	Months to years
Cost (for software + expertise)	Just your time	\$100s → \$1000s	~\$15 k → \$100 k



Your assignment based at this level

Classification: In-Confidence

Practicalities of LCA

Streamlined studies (like in your assignment)

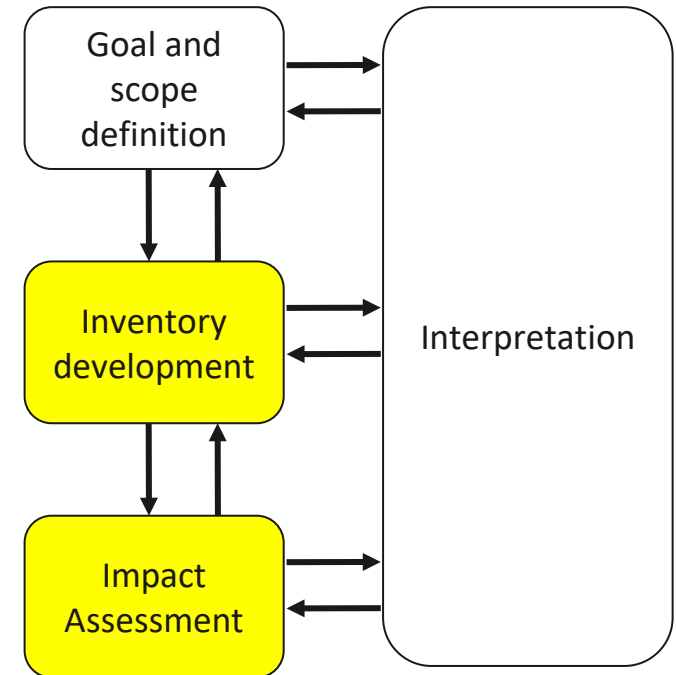
$$E = Q \times F$$

E = Emissions from the emissions source in kg CO₂-eq

Q = Activity data, e.g. quantity of material used

F = Emission factor for emissions source

1. Set functional unit, system boundary
2. Collect data
3. Use $E = Q \times F$ in a spreadsheet, add everything up
4. Produce a report



Source: ISO 14040:2006

Take-aways

1. Goal and scope definition
 2. Inventory development
 3. Impact Assessment
 4. Interpretation
- Getting the basics right is critical; goal and scope definition, and inventory data
 - Streamlined LCA uses $E = Q \times F$, to minimise complexity and time, but at the expense of certainty, resolution

What's next?

Week	Lectures Tuesday 1-2 & Friday 10-11	Tutorial / Workshop Weeks 7-12 Wednesday 3-4 or 4-5
7	Life Cycle Assessment (2/2)	Life Cycle Assessment
	Social and Economic Assessment	
8	Time Value of Money & Cash Flow	Time Value of Money & Cash Flow
	Cost Accounting	
9	Financial Accounting & Reporting (1/2)	Cost Accounting, Financial Accounting, Financial Reporting
	Financial Accounting & Reporting (2/2)	
10	Project Management	Project Management
	Risk Management	
11	Opportunity Analysis	Support for Sustainability Assignment
	Integrated Decisions (1/2) Guest Lecturer – Mike Ullrich (Fisher & Pykel Technologies)	
12	Integrated Decisions (2/2)	Risk Management and Opportunity Analysis
	Course Summary	

Details of the test...

- Tuesday 08 September 18:30 – 19:30
- Write-on test; GradeScope will be used to expediate grading
- No cheat-sheets or calculators allowed
- Five short-answer questions:
 - Bias, diversity and ethics
 - Law in New Zealand
 - Intellectual Property (more weighting on this question)
 - Business Structures, Strategy, Decisions and Risk
 - Data Privacy and Data Ethics
- Best preparation: review all content, including homework readings / assigned tasks
- Previous tests/exams; note limitations to these